MILITARY SPECIFICATION

FELT, BALLISTIC, NYLON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of needle-punched nylon felt resistant to penetration by ammunition fragments.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

A-A-203	- Paper, Kraft, Untreated
T-T-871	- Twine, Cotton, Wrapping

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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T-T-911	- Twine, Fibrous, Jute
CCC-C-429	- Cloth, Osnaburg, Cotton
CCC-C-467	- Cloth, Burlap, Jute (or Kenaf)
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner
PPP-T-45	- Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

MILITARY

MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible
MIL-T-40625	- Tubing, Bias Sewn (Burlap or Osnaburg), Cloth
MIL-P-46593	- Projectile, Calibers .22, .30, .50, and 20 mm Fragment-Simulating

STANDARDS

FEDERAL

FED-STD-123	- Marking for Shipment (Civil Agencies)
FED-STD-191	- Textile Test Methods

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-662	- Ballistic Test for Armor

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Rules and Regulations Under the Textile Fiber Products Identification Act

(Application for copies should be addressed to the Federal Trade Commission, Washington, DC 20580.)

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* 2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Material.

3.2.1 Construction. The felt shall be a nonwoven, needle-punched construction and shall be made with not less than 97 percent nylon when tested as specified in 4.4.3.

* 3.2.1.1 Nylon. The nylon shall be bright, high tenacity, 6.0 ± 1.5 denier per filament, 3.0 inch minimum to 4.0 inch maximum staple length, and crimped. The denier and staple length values apply to the fiber before crimping. The use of reprocessed nylon and mixtures of nylon fiber types shall be prohibited. The contractor shall identify the type of nylon used and the nylon manufacturer's commercial fiber designation.

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3.3 Color. The color of the finished felt shall be natural white.

3.4 Finish. The felt shall be needled and pressed so as to meet the physical requirements (see 3.5) and the ballistic resistance requirements (see 3.6) of this document.

* 3.5 Physical requirements. The finished felt shall weigh 51 ± 3 ounces per square yard and shall be 0.40 ± 0.05 inch in thickness when tested as specified in 4.4.3.

3.5.1 Width. The finished width of the felt shall be as specified (see 6.2).

3.6 Ballistic resistance. The ballistic limit V_{50} for the felt (one thickness) shall be not less than 1020 feet per second and no single ballistic determination on a sample unit shall be less than 980 feet per second when tested as specified in 4.4.3.

3.7 Length and put-up. Unless otherwise specified (see 6.2), the felt shall be put up on rolls in continuous lengths, and each roll shall be not less than 25 nor more than 30 yards in length.

3.8 Identification tickets. Unless otherwise specified (see 6.2), each roll of felt shall have a ticket attached to the edge at the inner end of the felt with not finer than 5-ply cotton string doubled to not less than 8 inches long. The ticket shall be made of not less than 20-point paper stock, and the color shall be white or light in intensity to permit easy reading of printed, stamped, or typed markings. Entries on tickets shall be printed, stamped, or typed. Handwritten entries are prohibited. The ticket shall have a reinforced eyelet for attaching the tying cord and shall be legibly printed with water-insoluble ink with the following information:

Item description
Federal Stock Number
Nylon Type
Supplier
Contract number and date
Document number and date
Name of contracting office
Contractor's piece number
Net yards

3.9 Fiber identification. Each roll of felt shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

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3.10 Workmanship. The finished felt shall conform to the quality and grade of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 3.1), it shall be visually examined for appearance and color, and tested for the characteristics listed in table I. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Component certification. The contractor shall submit a certificate of compliance for the characteristics listed below. When required, the certificate shall be accompanied by actual test, inspection, or other verifiable quality data.

Characteristic	Requirement paragraph
Nylon:	
Identification type	3.2.1.1
Reprocessed nylon prohibited	3.2.1.1
Mixtures of nylon fiber types prohibited	3.2.1.1
Tenacity	3.2.1.1
Luster	3.2.1.1
Crimp	3.2.1.1
Denier	3.2.1.1
Staple length (uncrimped)	3.2.1.1
Producers commercial fiber designation	3.2.1.1

4.4.2 End item examination.

4.4.2.1 Yard-by-yard examination.

4.4.2.1.1 Sampling examination. The required yardage of each roll shall be examined on one side only for the defects listed below. However, the side to be examined shall be alternated for each roll. A continuous defect shall be counted as one defect for each lengthwise yard or fraction thereof in which it occurs. The lot size shall be expressed in units of yards. The sample unit shall be 1 linear yard of felt. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5.

Defects

Any spot, stain, or streak 1/
Crease or wrinkle, hard embedded 1/

1/ Clearly visible at normal inspection distance (approximately 3 feet).

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4.4.2.1.2 One hundred percent examination. A 100 percent examination shall be conducted on both sides of every roll for defects of holes, cuts, tears, thin spots, thin edges, and broken needles embedded in the felt. Any roll containing one or more of these defects shall not be accepted.

4.4.2.1.3 Overall examination. Each defect listed below shall be counted not more than once in each roll examined. The sample size (number of rolls selected as sample) shall be those chosen for sampling examination (see 4.4.2.1.1). The lot shall not be considered acceptable if one or more defects are found in the sample.

Overall defects

Width less than specified
Objectionable odor
Fiber identification marking omitted from any roll

4.4.2.2 Length examination. During the sampling examination in 4.4.2.1.1, each roll in the sample shall be examined for length. Any length found to be less than the minimum or more than the maximum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective with respect to length or if the total of the actual lengths of the rolls in the sample is less than the total of the lengths marked on the tickets.

4.4.3 End item testing. The methods of testing specified in FED-STD-191 wherever applicable and as specified in table I shall be followed. Except for ballistic resistance, the physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. For the ballistic resistance characteristic, the lot average of all sample units in the sample shall apply. Except for ballistic resistance, the lot shall be unacceptable if one or more sample units fail to meet any requirement specified. For ballistic resistance, the lot shall be unacceptable if the lot average fails to meet the minimum ballistic resistance requirements or if any single determination falls below 980 feet per second (see 3.6). The sample unit for test purposes shall be as follows:

a. For ballistic tests, one cut 2/3 yard long and full width of the finished felt.

b. For all other tests, one cut 1/3 yard long and full width of the finished felt originating from one of the pieces from which the sample unit of the ballistic test is drawn.

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Each individual cut comprising sample unit (a) and (b) shall be marked to indicate contractor's piece, lot, and roll number and the Government lot number. The sample size shall be as shown below:

Lot size (yards)	Sample size (sample units)
Under 1,200	5
1,201 up to and including 35,000	8
35,001 and over	13

TABLE I. End item tests

Characteristic	Requirement paragraph	Test method
Nylon content 1/	3.2.1	2530
Weight	3.5	5041
Thickness	3.5	5030
Ballistic resistance	3.6	4.5.1

1/ The contractor shall submit a certificate of compliance for this characteristic. When required, the certificate shall be accompanied by actual test, inspection, or other verifiable quality data.

4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

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Examine	Defect
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.5 Methods of inspection.

4.5.1 Ballistic resistance test. The test shall be conducted in accordance with MIL-STD-662 and as specified below.

4.5.1.1 Test panel. The cut shall be held firmly by clamping in the width direction in a target holder. There shall be a minimum area of 8 by 12 inches of unsupported felt between supports. After this area has been tested, the cut shall be moved to permit the test to be continued on a fresh area. If desired, the felt may be cut into shorter length panels than its full width to facilitate mounting. Prior to conducting the ballistic test, the cut or test panel shall be preconditioned in the ballistic test area for at least 48 hours with air freely circulating on all sides of the panels.

4.5.1.2 Test projectile. The test projectile shall be the caliber .22 type 2 Projectile for Body Armor as specified in MIL-P-46593.

4.5.1.3 Fair impact. An impact is considered fair when an unyawed fragment simulator test projectile strikes the armor test panel at least 4 inches from any previous impact or at least 2 inches from a supported area or free edge of the test panel.

4.5.1.4 Calculation. The V_{50} ballistic limit (protection) for each test panel shall be the average of the velocities recorded for ten fair impacts consisting of five lowest velocities recorded for complete penetration and of the five highest velocities recorded for partial penetration, provided the spread for the ten velocities used is not greater than 125 feet per second. In cases where the spread is greater than 125 feet per second, the V_{50} limit shall be the average of fourteen fair impact velocities consisting of the seven lowest complete penetration velocities and of the seven highest partial penetration velocities. All velocities used in these calculations shall be

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striking velocities calculated at the point of initial contact with the test panel. The drag coefficient, K_D, to be used for making this calculation in conformance with the method of MIL-STD-662 shall be taken from table II.

TABLE II. Drag coefficient for striking velocities

Velocity (ft/sec)	Drag coefficient (K_D)
876 - 925	.0036
926 - 975	.0040
976 - 1025	.0042
1026 - 1075	.0044
1076 - 1125	.0047
1126 - 1175	.0050
1176 - 1225	.0052

4.5.1.5 Specimens. Three specimens shall be tested per sample unit.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

* 5.1.1 Level A. Felt shall be wound open width on convolute chipboard tubes. The ends of the tube shall be flush with, or extend not more than 1 inch beyond, each side of the width of the rolled felt. Each roll shall be completely enclosed in two thicknesses of 60-pound basis weight kraft paper conforming to A-A-203. The paper wrapping shall be secured with 2-inch minimum width gummed paper tape conforming to class 3 of PPP-T-45.

* 5.1.2 Commercial. Felt shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

* 5.2.1 Level A packing. Each roll of felt, preserved as specified in 5.1, shall be completely overwrapped in a waterproofed flexible sheet conforming to type III, grade B or C of MIL-L-10547. The seams and folds of the overwrap sheet shall be secured by one of the methods specified in the appendix of MIL-L-10547. Each overwrapped roll of felt shall be wrapped and sewn in

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cotton osnaburg cloth conforming to class 2 of CCC-C-429, or jute (or kenaf) burlap cloth conforming to class 1 of CCC-C-467. The wrapping shall be securely sewn with 16-ply cotton twine conforming to T-T-871 or jute twine conforming to type I, size 2 of T-T-911, with approximately one stitch per inch and every third stitch knotted. Alternatively, the seams of the wrap may be machine sewn. In lieu of the sewn wrap, bias-sewn tubing conforming to MIL-T-40625 may be used. A minimum of 5 inches of wrapping material or tubing shall be gathered together on each of the four corners and securely sewn into ears for handles. Wherever machine sewing is used, the thread type, stitch type, and number of stitches per inch shall be in accordance with MIL-T-40625. Alternatively, shipping containers shall be overseas type of PPP-B-601 or class 2, style 2 or 4 of PPP-B-621. Each shipping container shall be closed and strapped in accordance with the applicable container document. The weight of rolls and bundled sheets of felt in a cloth covering or shipping container shall not exceed 200 pounds, except where the prescribed length allowance of an individual roll exceeds this weight.

* 5.2.2 Level B packing. Each roll or bundle of felt, preserved as specified in 5.1, shall be wrapped in cloth covering as specified in 5.2.1. The waterproof barrier-material overwrap shall not be required. Alternatively, shipping containers that conform to domestic type, style A or B of PPP-B-601 or class 1, style 1 or 4 of PPP-B-621 shall be used. The weight of rolls and bundled sheets of felt in a cloth covering or shipping container shall not exceed 200 pounds, except where the prescribed length allowance of an individual roll exceeds this weight.

* 5.2.3 Commercial packing. The felt, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking.

* 5.3.1 Civil agencies. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in accordance with FED-STD-123 or ASTM D 3951, as applicable.

* 5.3.2 Military requirements. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The felt is intended for use in the manufacture of armor, including that worn by personnel, to provide protection against ammunition fragments.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Width required (see 3.5.1).
d. Length required, if other than specified (see 3.7).
e. Identification required, if other than specified (see 3.8).
f. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Felt processing information. A total apex angle of not more than 10° (or 5° if measured from perpendicular) in batt formation has appeared advantageous. Needling combinations of individual lightweight batts that have been needled about 277 penetrations per square inch, approximately 5/8 inch needle penetration as few times as possible is preferable to needling the combination each time a batt is added. Satisfactory felts have been made by needling each batt separately about 277 penetrations per square inch with 5/8 inch needle penetration and then needling all batts together with one needling or needling half the batts together then needling the halves together. If it is necessary to make the batts more compact before the final needling, a tacker or equivalent has been used at 40 to 60 penetrations per square inch. Needled thickness of the felt is about 5/8 inch. Pressing near the maximum thickness permitted is beneficial rather than pressing near the minimum thickness.

* 6.5 Subject term (key word) listing.

Armor
Ballistic protection
Felt
Nylon

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Custodians:

Army - GL
Air Force - 99

Preparing activity:

Army - GL

Project No. 8305-0096

Review activities:

Army - MD
Air Force - 11, 82
DLA - CT

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-F-43539A	Felt, Ballistic, Nylon

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6, REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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